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Developing Trends and Challenges of Digital Forensics

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***Abstract* - Digital forensics is concerned with identifying, reporting and responding to security breaches. It is about how to acquire, analyze and report digital evidence and using the technical skills, discovering the traces of Cyber Crime. The field of digital forensics is in high demand due to the constant threats of data breaches and information hacks. Digital Forensics is utilized in the identification and elimination of crimes in any controversy where evidence is preserved in online space. This is the use of specialized techniques for retrieval, authentication and electronic data analysis. Computer forensics deals with the identification, preservation, analysis, documentation and presentation of digital evidence. The paper has analyzed the present-day trends that includes IoT forensics, cloud forensics, network forensics and social media forensics. Recent researches have shown a wide range of threats and cyber-attacks, which requires forensic investigators and forensics scientists to simplify the digital world. Hence, all our research gives a clear view of digital forensics which could be of a great help in forensic investigation. In this research paper we have discussed about the need and way to preserve the digital evidence, so that it is not compromised at any point in time and an unalter evidence can be presented before the court of law.**

Keywords: Digital forensics, Cyber Crime, Digital Evidence, Social media forensics, IoT forensics, Cloud forensics, network forensics.

I. INTRODUCTION

Over the last few years, as computers are being connected from minute local networks to world's largest network i.e., the Internet, the term digital forensics has been used to cover the entire field.

The field of digital forensic includes network forensic also, and also includes the areas of expertise like investigation of breaches in network security, attempts to hack and data compromise [1]. Digital forensics comprises collection, review, evaluation, and thereafter recording artifacts and the events that are recorded as per sequence so that it can be presented as the proof of crime in the courtroom. Harmful methods, tools, and software are used and designed on a daily basis to harm the public and private networks while simultaneously using data storage, by extracting useful information. In a social network powered world, the enhancement of technology comes with the extension that is cyber-crime, which continues to creates new types of threats, attacks, tools, and tactics which allows the attackers to find a loophole in more complex or well-controlled environments, causing more damage, and sometimes remain unpredictable. As digital forensics techniques are getting more advance, hackers are equally smart they use anti-forensics tactics to generate delays or completely delete the digital evidence [2,3]. This study is devoted to analyze the threats, opportunities, challenges, and developing trends of digital forensics in cyber-sec department.

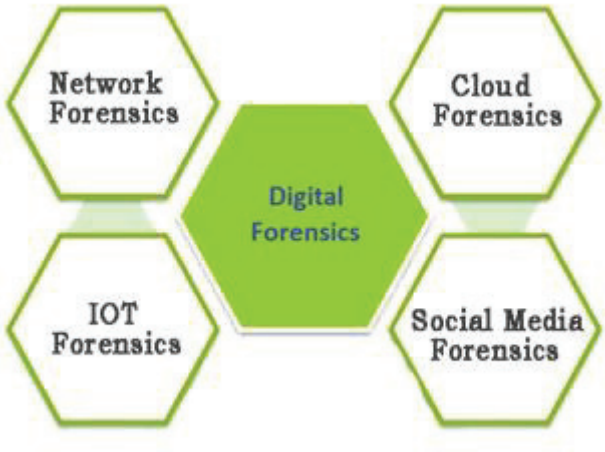

Fig. 1. Types of digital forensics

II. NEW EDGES IN DIGITAL FORENSICS

In this segment we have provided a short analysis of the types and trends of digital forensics.

A. Network Forensics:

Network forensics is to record, store and analyze network traffic. It is a critical branch of digital forensics which relates to computer network traffic monitoring and analyzing it for data collection, legal artifacts, or detecting intrusion. Network traffic is communicated and then lost, so it is often a pro-active investigation. Network forensics examiners established must establish standard procedures for how to acquire data after an attack or intrusion. Procedures must be based on the need of organizations and complement the network infrastructures. The forensic network solution records network traffic, stores it in searchable storage space and provides IT developers filters for extracting database to

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detect and analyze network abnormalities. Using network forensics, IT engineers can determine both the reasons of the abnormalities and its consequences on IT services and IT assets such as databases and servers. Sets up layer of protection to hide the most valuable data in an innermost layer of network.

Challenges:

The major challenge in network forensics is its large quantity of data that network produces which is about gigabytes in a day. It becomes a very tedious task to find out evidence, if the incident is explored after a long time. Another challenge in network forensics is in the unpredictability of Internet protocols. Some or the other way of addressing the 'to' and 'from' points is utilized by each network layer, e.g.: - Physical addresses (MAC Address), IP Addresses and Mail addresses (E-mail Addresses), which all can be deleted or modified(altered). Fortunately, a range of heavy software, including the products which are designed for the purpose of forensic analysis, makes it easier to resolve the cases by network work analysis.

Eventually the goal of network forensics is providing adequate evidences that permit the criminal culprit to get successfully summon. The effective use of Network Forensics could be in the areas including hacking, fraud analysis, insurance companies, data theft-industrial espionage, contempt, cloning credit card, software piracy, electoral law, indecent publication, false evidences, murder, sexual harassment, and discrimination on different basis.

B. IoT Forensics

Internet of Things (IoT) is a new paragon that is clearly increasing its field very fast in the present scenario of telecommunications. Desktops, laptops, tablets, smart devices, Personal Digital Assistance Devices, are majorly referred as IoT devices.

IoT forensics is a part of digital forensics which comprises of the cyber-crime that are done through IoT-devices and includes investigation of those devices, related sensors and the data that is stored on possible platforms. Significant advantage or benefits are provided in both business and domestic applications by using connectivity and intelligence of IoT systems[4]. However, we have several advantages but, IoT systems are prone to a vast number of security threats and attacks, like mass surveillance, IoT networks destruction, ransomware and Denial of Service (DoS). A growing number of IoT devices in private spaces such as smart homes pose opportunities and risks from a forensic perspective[5]. The evidence which are generated by the devices can be used for investigation and forensic analysis at the time of offence. Device level forensic includes the potential digital evidence that is collected from the IoT devices by images, audio, video. Other examples of device level forensics can be videos from CCTV camera and audios from Google Home or Alexa like cloud-based audio services. A richer and authentic source of evidence can be given by a IOT device, when compared with normal computer systems. IoT device forensics responds to the needs of users without requiring user interaction. As a result, the environment of IoT forensics provides circumstantial evidence that is of help to digital forensic researchers in analyzing visual world events.

Challenges:

Variety of challenges are included in the investigation of IoT devices that are based on non-compliance with the computer and traditional forensics methods presently available in IoT settings [6]. Many other factors, like restricted access to the crime scenes, service-level contracts limitation, dynamic architecture of computing, and standardization of hardware and software, makes IoT forensics indeed challenging.

Therefore, IoT forensics is one of the leading forms of digital forensics due to its capacity to give us proper content and digital evidences and because of the several challenges that are faced in this field.

C. Cloud Forensics:

Cloud forensics is the application of scientific principles, practices, and methods for organizing events by identifying, collecting, storing, testing, and reporting digital evidence. Cloud forensics has immersed major attention by the forensics experts because of the fact that massive resource pool is offered by cloud computing, economical solution, dynamicity, and wide access for storage. It is the critical application in digital forensics in a cloud-computing based environment. The prime attributes of cloud computing have remarkably decreased the costs of internet technology, rising the acceptability of cloud forensics all over the world by a large number of private organization and governments.

The high level of visibility, data duplication, power, and multi-tasking increased a variety of complex layers in cloud forensics. Despite the affirmative approach of the speedy development of cloud computing on the clients, amplified number of clients have also recognized the cloud atmosphere as one of the malicious fields [7]. Although difficulties are engrossed with cloud forensics, Cloud forensics is irrevocable that the turning up of cloud computing has increased confidentiality and availability issues.

Thus, it can be proved that it is among the significant trends that digital forensic had. This is due to the reason that it allows the forensics investigators in taking complete benefit of cloud computing attributes, which includes distributed forensic processing, computing power, reporting and scalability.

Challenges:

- Deleted information from crime scene investigation perspective, modified information and crediting the modified information to a particular client are important sources of verification.
- Decentralized data - Cloud computing permits data generation, storage, processing and distribution over different data centers, storage devices and physical machines.
- Data reliability guarantees the confirmation that exact representation of the information is available on the PC framework.

D. Social Media Forensics:

Social Media Forensic is a segment of the Network Forensics [6]. Social media forensics include the use of cyber research and digital analytics techniques for: Collecting information on social media platforms such as Facebook, Instagram, Twitter, LinkedIn, etc. Social networking attacks can occur within a network / program or offline. In social media forensics, a DoS attack can be described as an attack

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designed to prevent some media(social) services or resources from providing its normal services for a period of time. The criminal investigation is carried out with the help of Social Media forensics experts[8,9].

Forensic investigators can conduct effective investigations and gather convincing evidence legally if they are provided with sophisticated tools to manage the variability and size of social media content [10].

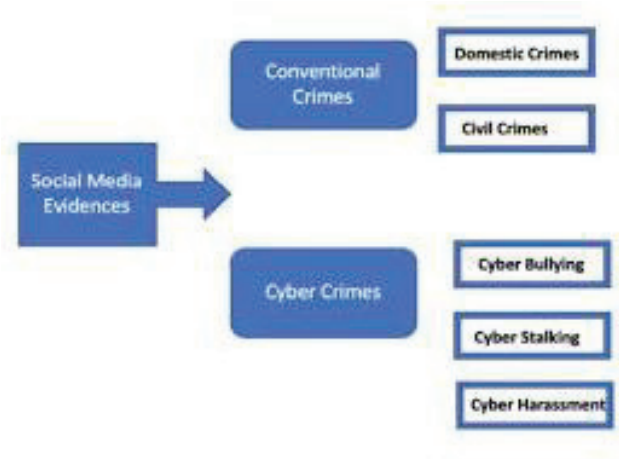

Fig. 2. Social media evidence in traditional and Cyber Crimes.

III. Idea Proposed and Implementation:

In today's time as we see that the security and availability of data is an important concern for everyone, same goes with the forensic data and evidence. After the digital evidence is captured, the concern is about how to preserve it securely so that at the time of presentation of evidence in court it is easily available unaltered and one way to achieve this is, by the means of using cloud as a storage option, any digital evidence when captured if at that time its backup is stored on cloud storage then it would give an extra level of security to the evidence.

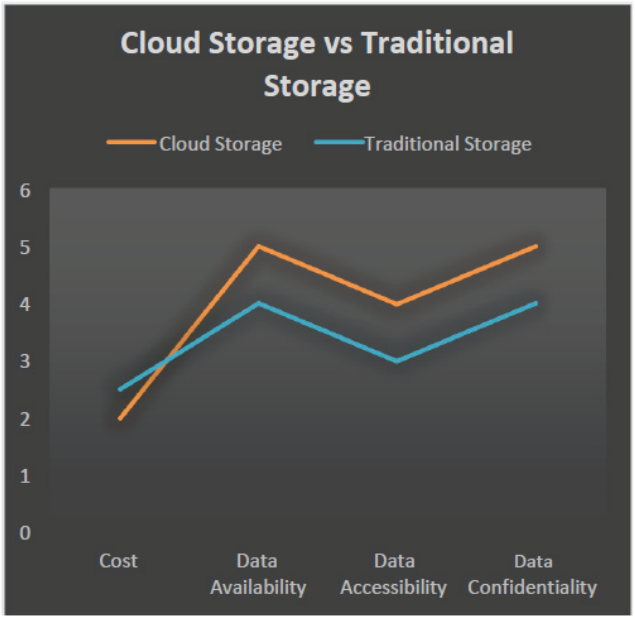

Fig. 3. Comparison cloud storage and traditional storage.

Using cloud as a solution to store digital evidence is advantageous as it improves the confidentiality, integrity and availability of the digital evidence, also using cloud storage is cheaper than using external drive for storage therefore the cutting the storage cost [11,12]. Talking about security we can implement logical access control to ensure that only the allowed person can access the evidence data, stopping unauthorized access, even if the username or password is compromised two-factor authentication can be used [13]. Also, robust security measures and be deployed over the cloud which includes strong encryption and advanced firewalls protecting the evidence from being tampered [14]. Regular monitoring can also be done so that if any suspicious activity is noticed the required action can be taken immediately.

Cloud storage offers the best security, performance and flexibility to extend storage on demand. It is available online so it can be used almost anywhere, on any device and for multiple users. However, it requires an ongoing monthly payment, depending on the amount of cloud storage you required to store the data.

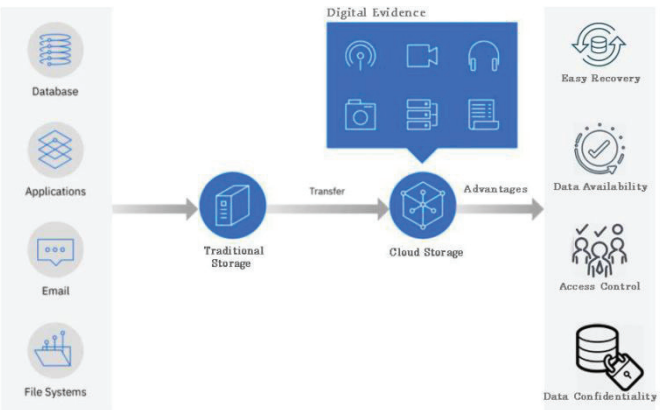

Fig. 4. Flow chart showing advantages of cloud storage

A. Achieving Data Confidentiality

In cloud data confidentiality can be achieved by providing access control management by proving the access to critical files only to those who are authorised for that, categorize sensitive data and provide access control using that. Can implement encryption of critical files so that by any chance data is compromised it is of no use for the attacker.

B. Achieving Data Availability

Data availability is a serious concern in the evidence presentation even after the evidence is captured by the authorities and stored in physical storage devices, but at the time of evidence presentation it remains unavailable it can be by the means of data mishandling or device damage. But with the use of cloud this issue will not occur as the data will be available all the time to the authorized personnel.

C. Cost

As the amount of data increases the problem to store it increases same comes with cost storing a huge amount of data in physical storage device is tough and much more expensive as compared to that of cloud, if the amount of data is less than physical devices can be a better option amount of data increases cloud becomes a better option.

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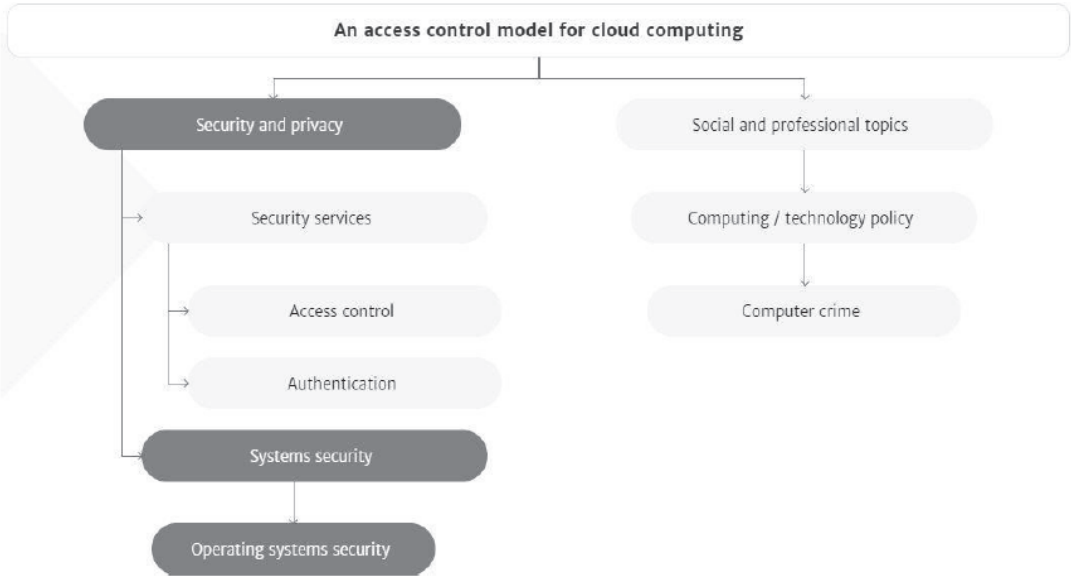

Fig. 5. Access Control Model for Cloud

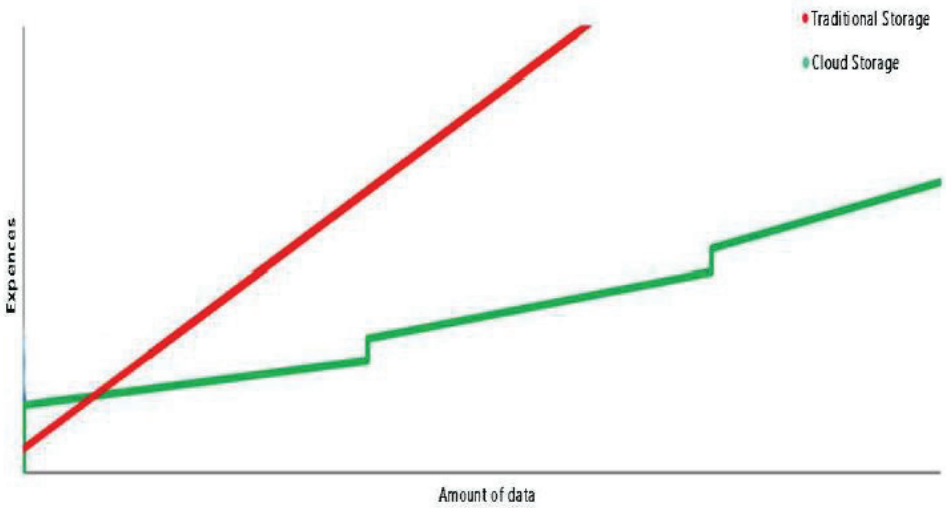

Fig. 6. Cost comparison cloud vs traditional storage

IV. Conclusion:

Digital forensics has received significant attention because cyber-crimes are increasing at a high rate. This paper discusses a short encounter with digital forensics, and the perpetuation of present trends that are observed by a number of experts. We have shown that if cloud storage is used instead of traditional storage methods in digital forensic then this will be much more advantageous, as problems of cost effectiveness, data confidentiality, integrity, accessibility and availability can be resolved to an extent. Also discussed about how these applications can be implemented. The future of digital forensic research is now concentrated on specific strengths that are now required to evolve. The paper also addresses the major difficulties encountered by the forensic experts. These security threats and breaches had prompted the research in digital forensics area showing that how digital evidence can be recovered from the forensic devices and further can be used for criminal and public legal proceedings. Investigators must be focused on advancing the approaches that are in line to a rising lawful framework and can develop the techniques by which the digital evidence that we store on cloud can reach to the court in an untouched way.

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